EXECUTIVE SUMMARY

hopLink

1 Application Overview

hopLink is a desktop application that scans an Excel workbook for every URL it contains, verifies each one against the live web, and produces an Excel report listing only the broken or potentially blocked links — including the exact cell location of each one in the original file.

The application replaces a manual process in which a project manager opens each link in a browser, waits for it to load, judges whether it works, and records the result by hand. On a workbook of several hundred URLs that process consumes a full day of focused work; on a workbook of several thousand it becomes a multi-week task. hopLink performs the same check in minutes.

2 Who It's For

PRIMARY USERS

Project managers, content owners, and operations staff responsible for spreadsheets containing web links — vendor directories, reference libraries, prior-authorization forms, compliance documentation, and similar workbooks.

TYPICAL USE CASES

- Quarterly or ad-hoc audits of large URL inventories
- Pre-publication validation of partner- or member-facing workbooks
- Periodic reverification of forms, microsites, and resource libraries

3 Key Benefits

- **Speed.** Reduces a multi-hour or multi-day manual task to minutes of computer time and roughly one minute of human attention per run.
- **Complete coverage.** Scans every column and every row automatically; URLs in non-obvious columns, embedded in descriptive text, or attached to hyperlinked cells are all detected.
- **Auditable output.** Every checked URL is recorded with HTTP status code, failure type, response time, and UTC timestamp.
- **Accurate diagnosis.** Failures are categorized — HTTP 404, DNS error, SSL error, timeout, or possibly blocked — so users can prioritize real broken links over false positives.
- **Cell-level traceability.** Each broken URL lists every cell where it appears, so corrections can be made at every occurrence.
- **No installation, no training.** Single executable; the workflow is drag file → click → save report.

4 Time Savings & Efficiency Metrics

> **Assumption:** Manual checking is estimated at **45 seconds per URL** — a conservative midpoint covering the time to click the link, load the page, judge whether it works, and record the result. Faster checks (30s) are possible for trivial pages; more thorough checks (60s+) are common where recordkeeping is rigorous.

Workbook	Manual time (45s/URL)	App run time	Time saved per run	Speed-up
10,000 URLs (test)	125 hrs (~15.6 workdays)	1 hr 23 min	~15.5 workdays	90×
5,000 URLs (test)	62.5 hrs (~7.8 workdays)	36 min	~7.7 workdays	104×
1,000 URLs (test)	12.5 hrs (~1.6 workdays)	7 min	~1.5 workdays	107×
588 URLs (real-world)	**7 hr 21 min**	**< 2 min**	**~7 hours**	**220×**

Absolute time savings grow with workbook size: a 10,000-URL audit reclaims roughly three weeks of project-manager time per run. The speed-up multiplier moderates on larger files because HTTP checking takes real network time, but the practical impact grows — manual processing becomes prohibitively expensive at scale.

5 System Location & Accessibility

- **Location:** Stored on the shared **X: drive → Apps** folder; accessible to anyone with shared-drive access.
- **Local execution:** Runs directly on the user's Windows machine. The executable launches from the shared drive — no copy step required.
- **No installation, no admin rights, no external dependencies.** Nothing is written to the registry. No Python, libraries, or supporting tools required.
- **Data stays local:** No data leaves the user's machine except for the HTTP requests the application makes to verify URLs.

BOTTOM LINE

hopLink turns a recurring multi-day manual chore into minutes of computer time, with a more complete and auditable result. Already built, on the shared drive — no installation needed.